

EXP.NO: 05	DESIGN OF FIR FILTERS USING MATLAB
Date:	

Aim:

To write a MATLAB program for the design of FIR filters (LPF/HPF/BSF/BPF) for the given cut off frequency using Rectangular, Hamming and Hanning windowing techniques. Also plot the magnitude and log magnitude responses for the same.

Apparatus Required:

HARD WARE: IBM PC (Or) Compatible PC

SOFTWARE: MATLAB 2015a (Or) Higher version **Algorithm:**

Step 1: Start the program.

Step 2: Get the cutoff frequency and sampling frequency from the user.

Step 3 : Get the filter order value of N.

Step 4: Get the window function $w(n)$ for different windowing techniques in Z-domain.

Step 5 : Design a linear phase FIR filter using 'fir1' command for LPF/HPF/BSF/BPF

Step 6: Compute the complex frequency response of digital transfer function at specified frequency points using 'freqz' command.

Step 7: Plot the magnitude response and log magnitude response.

Step 8: Terminate the process.

Program:

% Parameters

N = 7; % Filter length

wc = 1; % Cutoff frequency in rad/sample

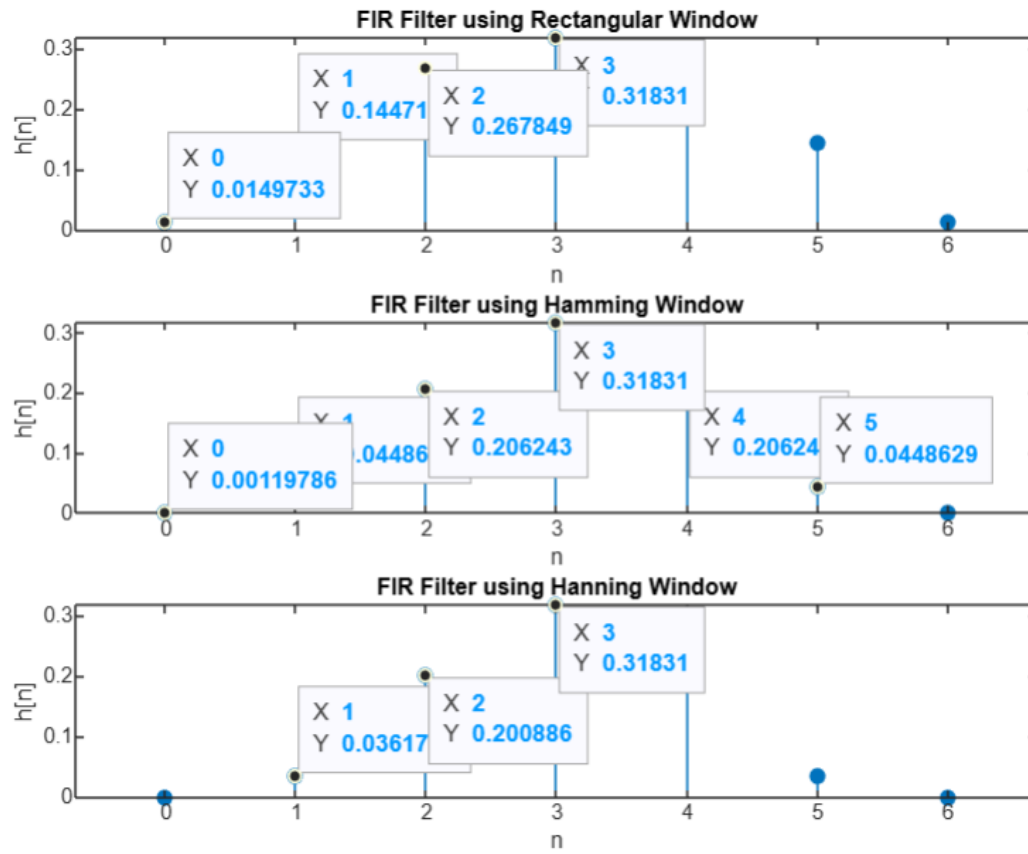
tau = (N - 1) / 2; % Group delay

n = 0:N-1; % Time indices

% Ideal impulse response hd[n]

hd = zeros(1, N);

OUTPUT :



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for i = 1:N
    if n(i) == tau
        hd(i) = wc / pi;
    else
        hd(i) = sin(wc * (n(i) - tau)) / (pi * (n(i) - tau));
    end
end

% Define Windows
rect_window = ones(1, N);
hamming_window = hamming(N)';
hanning_window = hann(N)';

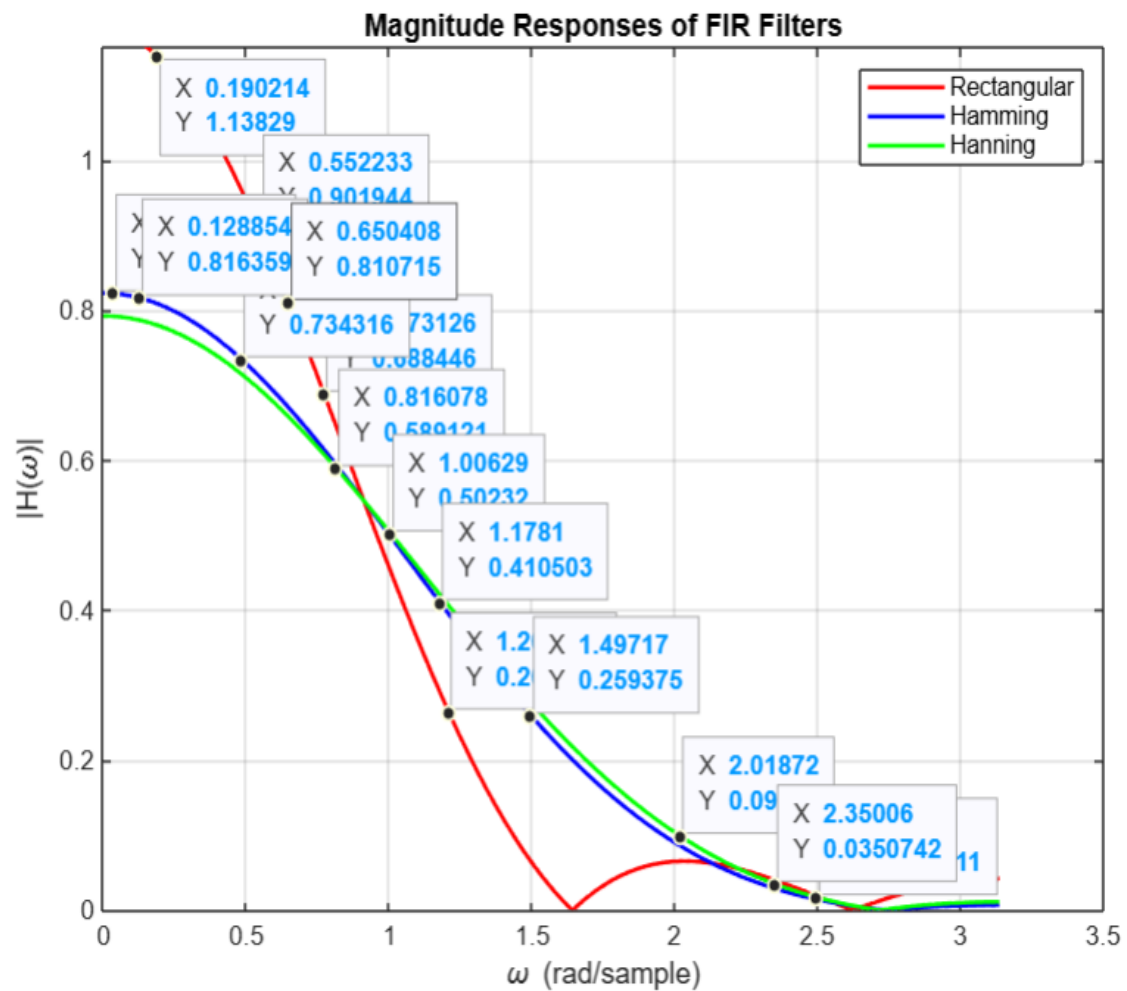
% Apply Windows
h_rect = hd .* rect_window;
h_hamming = hd .* hamming_window;
h_hanning = hd .* hanning_window;

% Plot Impulse Responses
figure;
subplot(3,1,1);
stem(n, h_rect, 'filled'); title('FIR Filter using Rectangular Window');
xlabel('n'); ylabel('h[n]');

subplot(3,1,2);
stem(n, h_hamming, 'filled'); title('FIR Filter using Hamming Window');
xlabel('n'); ylabel('h[n]');

subplot(3,1,3);
stem(n, h_hanning, 'filled'); title('FIR Filter using Hanning Window');
xlabel('n'); ylabel('h[n]');

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% Plot Frequency Responses
figure;
[H_rect, w] = freqz(h_rect, 1, 512);
[H_hamming, ~] = freqz(h_hamming, 1, 512);
[H_hanning, ~] = freqz(h_hanning, 1, 512);

plot(w, abs(H_rect), 'r', 'LineWidth', 1.5); hold on;
plot(w, abs(H_hamming), 'b', 'LineWidth', 1.5);
plot(w, abs(H_hanning), 'g', 'LineWidth', 1.5);
legend('Rectangular', 'Hamming', 'Hanning');
xlabel('\omega (rad/sample)'); ylabel('|H(\omega)|');
title('Magnitude Responses of FIR Filters');
grid on;
```


INFERENCE:

The design and implementation of FIR (Finite Impulse Response) filters using MATLAB have demonstrated the effectiveness and flexibility of digital signal processing techniques. MATLAB provides a comprehensive environment for filter design, enabling the use of various window methods such as Hamming, Hanning, Blackman, and Kaiser, each offering specific trade-offs between main-lobe width and side-lobe attenuation.

RESULT:

Thus the MATLAB program for the design of FIR filters (LPF/HPF/BSF/BPF) for the given cut off frequency using Rectangular, Hamming and Hanning windowing techniques are done and the magnitudes are plotted .

